

Paper Manager pack — FAST (Oct 19)

Paper: FAST: Efficient Action Tokenization for Vision-Language-Action Models

arXiv: <https://arxiv.org/abs/2501.09747>

Site: <https://pi.website/research/fast> · HF `physical-intelligence/fast`

PDF: `/workspace/papers/fast-2501.09747.pdf`

Session: Oct 19 · due 9:30 AM · **Empiricist=2** (only filled seat); full five packs

Sources: Summarizer + Senior Research (`/workspace/fast-quiet/DELIVERABLE.md`)

1. CRITICAL MYTH BUST (say first)

Label	Meaning
FAST	Discrete DCT → BPE action tokenizer for AR VLAs
FM	Continuous flow-matching action generation (π_0 / $\pi_{0.5}$ deploy)
π_0-FAST	Discrete AR variant using this tokenizer
$\pi_{0.5}$	FAST for pretrain , FM for post-train/deploy — FAST does NOT replace FM

Naming hygiene: $\pi_0 \approx$ FM; π_0 -FAST $\approx$ discrete AR; $\pi_{0.5} \approx$ hybrid successor.

2. Paper summary

Compression-based **DCT + BPE** action tokenization unlocks high-Hz AR VLAs where naïve per-dim binning fails. Ships **FAST+** universal tokenizer (~1M 1s chunks). **π_0 -FAST** matches diffusion/FM π_0 on dexterous generalist tasks at **~5× less train compute**, but infer **~750 ms** vs FM **~100 ms**/chunk (4090). First zero-shot DROID language generalist in unseen rooms.

3. Keywords / pipeline

Quantile norm → per-dim **DCT** → scale γ (default **10**) → round → low-freq-first flatten → **BPE** vocab **1024** → overwrite VLM vocab slots. Typical **1 s** chunks; invertible decode.

4. When FAST helps vs FM wins

FAST helps	Continuous FM wins
High-Hz next-token prediction	Inference latency / smooth dynamics
Train compute / compression	Closed-loop control (fewer ms/chunk)
Denser chunks without token blowup	Real-time household deploy

$\pi_{0.5}$ recipe: FAST for efficient pretrain NTP → serve with continuous FM.

5. Role packs

Stakeholder (18+3)

Ship FAST+ for AR training / cheap pretrain; keep FM for realtime deploy. Do **not** delete FM because FAST exists. Risk: AR latency $\sim 7.5\times$ vs FM.

Scientific Reviewer (8+3)

Strengths: causal story; invertible compressor; honest latency; open HF. Weaknesses: figure-only scores; FAST+ mix leakage; AR vs FM not FLOP-matched; DROID 44 trials. Verdict: cite as discrete tokenizer unlock / π_0 -FAST — **not** as FM retirement.

Empiricist (11+3) — WEIGHT 2 · LONGEST

H / smoke

Priority	Experiment
P0 Y	HF FAST+ encode/decode; compression vs naïve bins; didactic AR toy across Hz
P1 Partial	VLA FT FAST+ vs bins; BPE on/off; γ sweep; **AR-FAST vs continuous FM** matched task
P2 Partial	fit() vs universal; $\pi_{0.5}$ hybrid echo
Defer	Full π_0 -FAST 10k h; real laundry hardware

Reproduce: `AutoProcessor.from_pretrained("physical-intelligence/fast", trust_remote_code=True);  $\gamma=10$ ;  $|V|=1024$ .`

Sanity Table I: Bridge~20, DROID~29, bussing~28, shirt~53 vs naïve 35/105/140/700.

Latency: ~100 ms FM vs ~750 ms AR-FAST. Env: /workspace/fast-quiet/env_outline/.

Archaeologist (11+3)

Priors: JPEG/DCT, BPE, RT-2/OpenVLA naïve bins, ACT/DP/ π_0 chunks, π_0 FM, VQ/FSQ.

Forward: FAST $\rightarrow$ π_0 -FAST $\rightarrow$ $\pi_{0.5}$ (FAST pre $\rightarrow$ FM) $\rightarrow$ KI $\rightarrow$ RTC $\rightarrow$ BEHAVIOR. Keep labels distinct.

Visionary (11+3)

Keep FM(+RTC) at deploy; keep FAST for pretrain/language/compression. Invest AR decode speed if discrete VLAs compete. FAST+ default for new morphologies. Skip Discord/Drive.

6. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	FAST for train / FM for deploy
Reviewer	8+3	Not FM retirement
Empiricist	**11+3**	**Tokenizer smoke + AR vs FM (HIGH)**
Archaeologist	11+3	Naming hygiene π_0 / π_0 -FAST / $\pi_{0.5}$
Visionary	11+3	Hybrid for 2026 BEHAVIOR

Spine: myth bust + DCT $\rightarrow$ BPE pipeline. PDF for Mac.

GitHub title: [ECE 605] - FAST Making Empiricist

Paths

Artifact	Path
This pack	`/workspace/fast-quiet/KANTA_PACK.md`
Senior Research	`/workspace/fast-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/fast-2501.09747-summary.md`
Paper PDF	`/workspace/papers/fast-2501.09747.pdf`